Week 3 Tutorial Exercises

Question 1 — LO 2.3

For each of the following, say whether it is true or false.

i) $\emptyset \in \emptyset$

ii) $\emptyset \in \{\emptyset\}$

iii) $0 \in \{\emptyset\}$

iv) $\{\emptyset\} = \{\{\{\}\}\}$

Solution:

i) False.

ii) True.

iii) False.

iv) False.

Question 2 — LO 2.4

For each of the following, say whether it is true or false.

i) $1 \subseteq \{1\}$

ii) $\{1\} \subseteq \{1\}$

iii) $\{1, 2\} \subseteq \{1, 2, \{1, 2, 3\}\}$

iv) $\{1, 2, 3\} \subseteq \{1, 2, \{1, 2, 3\}\}$

v) $\{\{1, 2, 3\}\} \subseteq \{1, 2, \{1, 2, 3\}\}$

vi) $\{2, 3\} \subseteq \{2, \{3, 2\}\}$

vii) $\{2, 3\} \subseteq \{\{2, \{3, 2\}\}\}$

viii) $\{2, 3\} \subseteq \{2, \{3, 2, 4\}\}$

ix) $\emptyset \subseteq \emptyset$

x) $\mathbb{Z} \subseteq \mathbb{N}$

Solution:

i) F Reason: 1 is not a set

ii) T Reason: every set is a subset of itself

iii) T

iv) F Reason: $3 \in \{1, 2, 3\}$ but $3 \notin \{1, 2, \{1, 2, 3\}\}$

v) T

vi) F Reason: $3 \in \{2, 3\}$ but $3 \notin \{2, \{3, 2\}\}$

vii) F Reason: Neither 2 nor 3 is a member of $\{\{2, \{3, 2\}\}\}$

viii) F Reason: $3 \in \{2, 3\}$ but $3 \notin \{2, \{3, 2, 4\}\}$

ix) T Reason: The empty set is a subset of every set

x) F Reason: e.g. $-1 \in \mathbb{Z}$ but $-1 \notin \mathbb{N}$

Question 3 LO 2.5

With $\mathcal{U} = \{x \in \mathbb{N} : 1 \leq x \leq 10\}$, $A = \{2, 3, 4, 5, 6\}$ and $B = \{2, 4, 6, 8\}$ calculate the following:

i) $A \cup B$

ii) $A \cap B$

iii) A^c

iv) B^c

v) $B - A$

vi) $A - B$

vii) $A^c \cap B^c$

viii) $A^c \cup B^c$

ix) $A \triangle B$

Solution:

i) $\{2, 3, 4, 5, 6, 8\}$

ii) $\{2, 4, 6\}$

iii) $\{1, 7, 8, 9, 10\}$

iv) $\{1, 3, 5, 7, 9, 10\}$

v) $\{8\}$

vi) $\{3, 5\}$

vii) $\{1, 7, 9, 10\}$

viii) $\{1, 3, 5, 7, 8, 9, 10\}$

ix) $\{3, 5, 8\}$

Question 4 LO 2.5

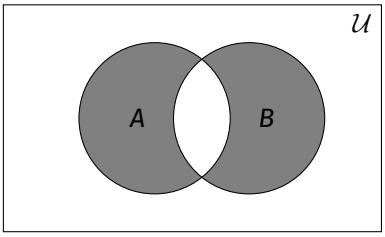
For any two non-empty sets A and B drawn from the same universal set $\mathcal{U}$, let $Z = A \triangle B$. Which of the following statements is true:

i) Z could never equal $\emptyset$

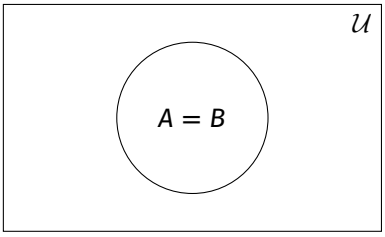
ii) Z could never equal $\mathcal{U}$

iii) Z could never equal B

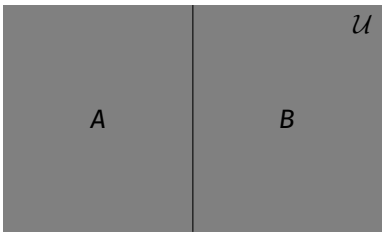
Solution: Consider first the general case for $Z = A \triangle B$.



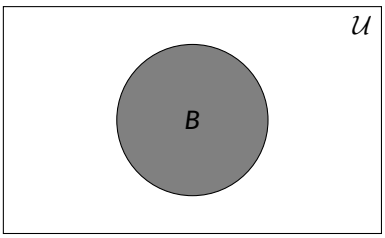
i) For Z to equal $\emptyset$, we must reduce the size of the shaded area to nothing. The size of the shaded area reduces as we move A over B, and when $A = B$ we have $Z = \emptyset$. So this is possible, so the statement is false.



ii) For Z to equal $\mathcal{U}$, we must reduce the non-shaded areas down to nothing. To do this, we must have $A \cap B = \emptyset$ and $A \cup B = \mathcal{U}$. So $Z = \mathcal{U}$ is possible, so the statement is false.



iii) For Z to equal B, the shaded region must be equal to B. To have all of B shaded, we must separate A and B completely so that they are disjoint. Then, to make the shaded region equal to B, we must reduce the size of A down to nothing, i.e. A must be the empty set. But we are told that A and B are both non-empty, so the statement that Z can never equal B is TRUE.

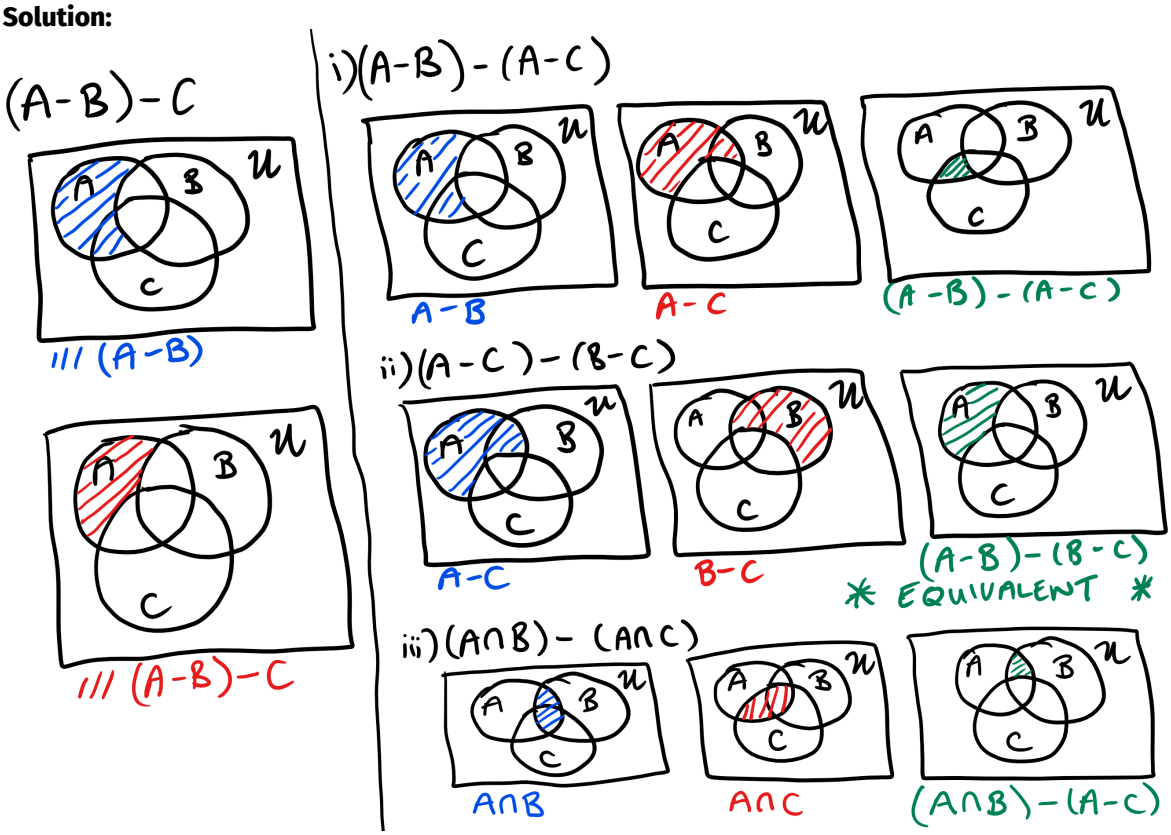


Question 5 LO 2.6

Let A, B and C be any three sets. Draw Venn diagrams to determine which of the following is equal to $(A - B) - C$:

i) $(A - B) - (A - C)$ **ii)** $(A - C) - (B - C)$ **iii)** $(A \cap B) - (A \cap C)$

Solution:



Answer: $(A - B) - C \equiv (A - C) - (B - C)$ [answer (ii)]

Question 6 LO 2.6

In a survey of 100 students:

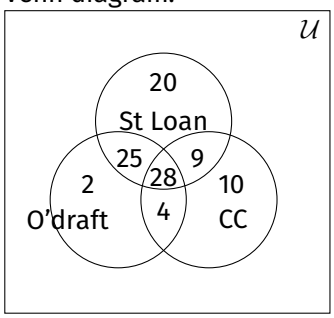
- **82 have a student loan.**
- **51 have a credit card.**
- **59 have an overdraft.**
- **37 have a student loan and a credit card.**
- **53 have a student loan and an overdraft.**
- **32 have a credit card and an overdraft.**
- **28 have a student loan, a credit card and an overdraft.**

i) Represent this information on a Venn diagram.

ii) How many of the students surveyed do not have a student loan, credit card or overdraft?

Solution:

i) Venn diagram:



ii) From Venn diagram: 100 - (28 + 4 + 25 + 9 + 2 + 10 + 20) = 100 - 98 = 2

Question 7 LO 2.7

Prove the following using the fundamental set identities:

i) $A - (B \cap C) \equiv (A - B) \cup (A - C)$

ii) $(A \cap B)^c \cap C \equiv (A^c \cap C) \cup (B^c \cap C)$

iii) $(A \cup B)^c \cup C \equiv (A^c \cup C) \cap (B^c \cup C)$

Solution:

i)
$$\begin{aligned} A - (B \cap C) &\equiv A \cap (B \cap C)^c && \text{by definition of '-'} \\ &\equiv A \cap (B^c \cup C^c) && \text{by de Morgan's laws} \\ &\equiv (A \cap B^c) \cup (A \cap C^c) && \text{by distributivity} \\ &\equiv (A - B) \cup (A - C) && \text{by definition of '-'} \end{aligned}$$

ii)
$$\begin{aligned} (A \cap B)^c \cap C &\equiv (A^c \cup B^c) \cap C && \text{by de Morgan's laws} \\ &\equiv (A^c \cap C) \cup (B^c \cap C) && \text{by distributivity} \end{aligned}$$

iii)
$$\begin{aligned} (A \cup B)^c \cup C &\equiv (A^c \cap B^c) \cup C && \text{by de Morgan's laws} \\ &\equiv (A^c \cup C) \cap (B^c \cup C) && \text{by distributivity} \end{aligned}$$

Question 8 — LO 2.8

What is the cardinality of the following sets:

i) $\{1, \{1\}, \{1, 1\}\}$

ii) $\{1, 1, 1, 2, 2, 2, 3\}$

iii) $\{x : x \text{ is a letter in the word CARDINALITY}\}$

iv) $\mathcal{P}(\{x \in \mathbb{N} : 1 \leq x \leq 32\})$

Solution:

i) 2

ii) 3

iii) 9

iv) $2^{32} = 4294967296$

Question 9 — LO 2.9

What is the power set of the following sets? Check their cardinalities using $|\mathcal{P}(A)| = 2^{|A|}$.

i) $\emptyset$

ii) $\{x\}$

iii) $\{x, y\}$

iv) $\{x, y, z\}$

Solution:

i) $\mathcal{P}(\emptyset) = \{\emptyset\}$ $\quad |P(\emptyset)| = 2^0 = 1$

ii) $\mathcal{P}(\{x\}) = \{\emptyset, \{x\}\}$ $\quad |P(\{x\})| = 2^1 = 2$

iii) $\mathcal{P}(\{x, y\}) = \{\emptyset, \{x\}, \{y\}, \{x, y\}\}$ $\quad |P(\{x, y\})| = 2^2 = 4$

iv) $\mathcal{P}(\{x, y, z\}) = \{\emptyset, \{x\}, \{y\}, \{z\}, \{x, y\}, \{x, z\}, \{y, z\}, \{x, y, z\}\}$ $\quad |P(\{x, y, z\})| = 2^3 = 8$

Question 10 — LO 2.10

If $A = \{a, b, c\}$ and $B = \{1, 2\}$, what is:

i) $A \times B$

ii) $B \times A$

iii) $B \times B$

iv) $B \times A \times B$

Solution:

i) $A \times B = \{(a, 1), (a, 2), (b, 1), (b, 2), (c, 1), (c, 2)\}$

ii) $B \times A = \{(1, a), (1, b), (1, c), (2, a), (2, b), (2, c)\}$

iii) $B \times B = \{(1, 1), (1, 2), (2, 1), (2, 2)\}$

iv) $\{(1, a, 1), (1, a, 2), (1, b, 1), (1, b, 2), (1, c, 1), (1, c, 2), (2, a, 1), (2, a, 2), (2, b, 1), (2, b, 2), (2, c, 1), (2, c, 2)\}$

Question 11 — LO 2.11

Let $S = \{1, 2, 3, 4\}$. Determine which of the following, if any, are partitions of S:

i) $\{\{1\}, \{2, 3\}\}$

ii) $\{\{1, 2, 3, 4\}\}$

iii) $\{\emptyset, \{1, 2\}, \{3, 4\}\}$

iv) $\{\{2\}, \{1, 3, 4\}\}$

Solution:

i) No: 4 is not covered.

ii) Yes.

iii) No: $\emptyset$ not allowed.

iv) Yes.